

Severe COPD Exacerbation with Pneumothorax

Section 1: Case Summary

Scenario Title:	Severe COPD Exacerbation with Pneumothorax
Keywords:	Hypoxia, dyspnea, pneumothorax
Brief Description of Case:	68M with respiratory distress. Known chronic obstructive pulmonary disease (COPD) on home O2, 3 days of worsening cough, sputum and dyspnea. Becomes increasingly distressed and agitated requiring intubation. Learners may consider non-invasive positive pressure ventilation (NIPPV) and delayed sequence intubation (DSI), but will require intubation. Shortly after intubation, becomes very difficult to bag and then becomes hypotensive. If not identified, the patient arrests from a tension pneumothorax.

Goals and Objectives	
Educational Goal:	Appropriately manage the patient with a severe COPD exacerbation, with consideration of delayed sequence intubation, and recognition/management of the potential complications of positive pressure ventilation (PPV) in this patient subset.
Objectives: (Medical and CRM)	1. Manage the patient with a severe COPD exacerbation, including intubation 2. Perform delayed sequence intubation 3. Trouble-shoot increased airway pressures/difficulty bagging post-intubation 4. Identify and treat a tension pneumothorax 5. Maintain situational awareness and control of the room when managing an agitated patient 6. Recognize a change in patient status
EPAs Assessed:	

Learners, Setting and Personnel				
Target Learners:	☐ Junior Learners		☒ Senior Learners	
	☐ Physicians		☐ Nurses	☐ RTs
	☐ Inter-professional			
	☐ Other Learners:			
Location:	☒ Sim Lab		☐ In Situ	
Recommended Number of Facilitators:	☐ Other:			
	Instructors: 1			
	Confederates: 1			
	Sim Techs: 1			

Scenario Development	
Date of Development:	14/03/2018
Scenario Developer(s):	Lindsey McMurray
Affiliations/Institutions(s):	The Ottawa Hospital
Contact E-mail:	mcmurray.lindsey@gmail.com
Last Revision Date:	07/06/2020
Revised By:	Beatrix Bérczi
Version Number:	2



Severe COPD Exacerbation with Pneumothorax

Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Peter Long		Age: 68		Gender: M	Weight: 60 kg
Presenting complaint: Shortness of Breath					
Temp: 37 °C	HR: 120/min	BP: 160/95	RR: 30/min	O ₂ Sat: 80%	FiO ₂ : 100% NRB
Cap glucose: 5 mmol/L			GCS: 11 (E4 V2 M5)		
Triage note: Patient has just arrived via EMS in severe respiratory distress. Patient is a known COPD patient on home O ₂ . MD called to the bedside.					
Allergies: NKDA					
Past Medical History: COPD (on home O ₂ , 2L) Hypertension Smoker			Current Medications: Tiotropium Salbutamol PRN Amlodipine		

Section 2B: Extra Patient Information

A. Further History	
<i>Include any relevant history not included in triage note above. What information will only be given to learners if they ask? Who will provide this information (mannequin's voice, confederate, SP, etc.)?</i>	
From home alone, no further history	

B. Physical Exam	
<i>List any pertinent positive and negative findings</i>	
General Status: Distressed, sitting up, respiratory distress	
Cardio: Normal heart sounds	Neuro: Eyes open, moaning intermittently
Resp: Diffuse wheeze/gasping	Head & Neck: Normal
Abdo: Normal	MSK/skin: Normal
Other: n/a	

Severe COPD Exacerbation with Pneumothorax

Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin: Adult
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Critical care supplies, Airway cart, NIPPV
C. Required Medications
Puffers/Aerochamber/Nebs Intubation Kit Arrest Kit
D. Moulage
None
E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed
☒ Patient not yet on monitor
F. Patient Reactions and Exam
<i>Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format.</i> Dyspnea, gasping Wheeze Moaning intermittently No verbal response

Severe COPD Exacerbation with Pneumothorax

Section 4: Confederates and Standardized Patients

Confederate and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)
Nurse	Skillful and cooperative bedside RN Prompts to patient's increasing agitation Prompts to consider Hs & Ts during cardiac arrest

Severe COPD Exacerbation with Pneumothorax

Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: sinus HR: 120 BP: 160/95 RR: 30 O ₂ SAT: 80% (neb) T: 37°C GCS: 11	Sitting up, distressed. Gasping for air.	<u>Expected Learner Actions</u> ☐ Don appropriate PPE (Airborne precautions for COVID) ☐ Focused history/physical ☐ Monitors/IVs ☐ 100% O ₂ (non-rebreather) ☐ ECG/Chest X-ray/Labs ☐ Steroids and antibiotics ☐ Bronchodilators ☐ NIPPV ☐ Attempt to clarify Code Status, ask for old chart	<u>Modifiers</u> - O ₂ increases to 85% with NRB <u>Triggers</u> <i>For progression to next state</i> - After 3-5 minutes → 2. Agitation	
2. Agitation HR: 130 RR: 30/min O ₂ SAT: 85% (NRB)	RN prompts that patient is more agitated and grabbing at face mask	<u>Expected Learner Actions</u> ☐ Consider delayed sequence intubation with ketamine ☐ BVM with PEEP valve ☐ Consider nasal prongs ☐ Post intubation sedation	<u>Modifiers</u> - With ketamine patient settles and can BVM with PEEP valve - Once intubated, O ₂ to 92%, HR 90 <u>Triggers</u> - 2 minutes post-intubation → 3. Pneumothorax	
3. Pneumothorax Rhythm: sinus HR: 130 BP: 70/50 RR: 16/min **no air entry Right chest** O ₂ SAT: 80% (Vent)	Patient develops obstructive shock from tension pneumothorax If patient arrests, RN may prompt learner to consider Hs & Ts of ACLS	<u>Expected Learner Actions</u> ☐ Troubleshoot high airway pressures. (Disconnect from vent, BVM, manually compress chest, suction, assess for pneumothorax) ☐ Needle decompress/finger thoracostomy one or both sides. ☐ Chest tube insertion ☐ Call ICU	<u>Modifiers</u> - Once pneumothorax is identified and decompressed, O ₂ and BP improve. <u>Triggers</u> - If pneumothorax not identified, after 2 minutes, patient has PEA arrest which will resolve with decompression.	



Severe COPD Exacerbation with Pneumothorax

Appendix A: Laboratory Results

<u>VBG</u> pH 7.15 pCO ₂ 80 pO ₂ 38 HCO ₃ 29 Lactate 4	
--	--

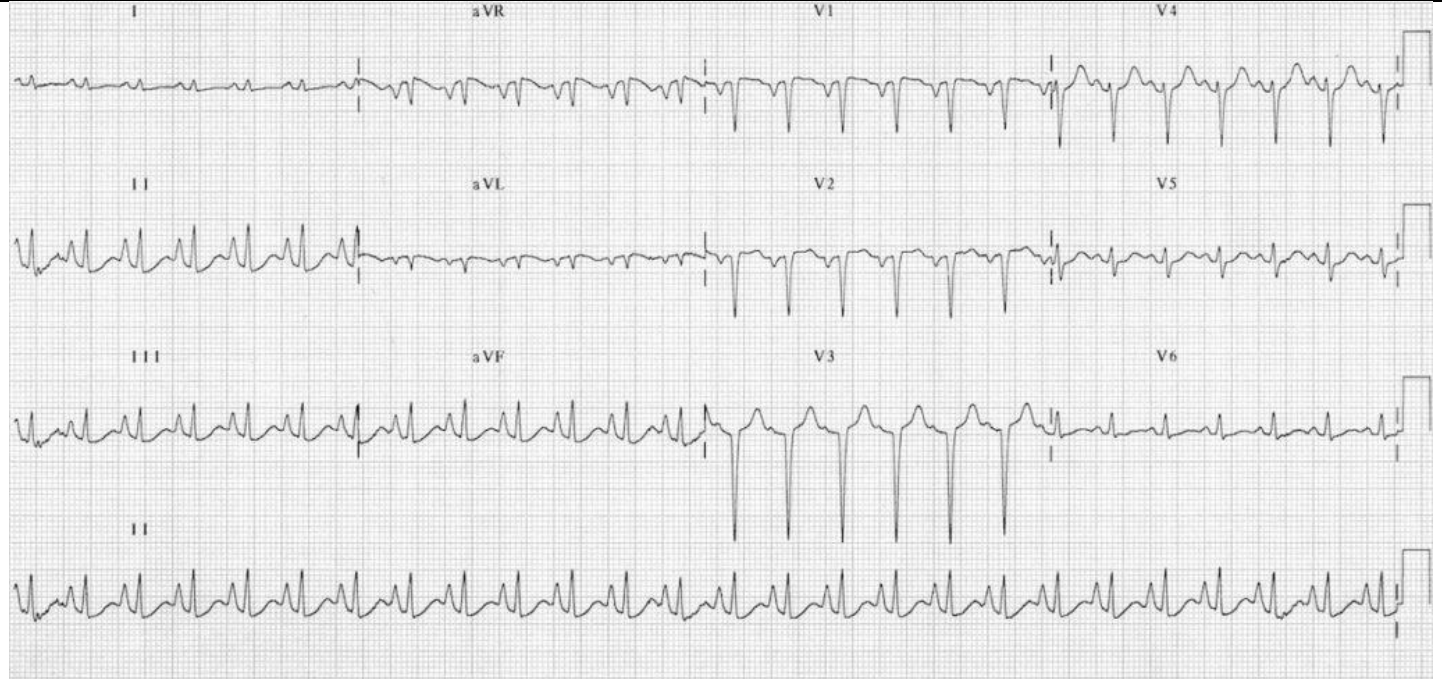
Appendix B: ECGs, X-rays, Ultrasounds and Pictures



Chest xray from radiopaedia.org

<https://radiopaedia.org/articles/chronic-obstructive-pulmonary-disease-1>

Severe COPD Exacerbation with Pneumothorax



ECG from LITFL

<https://litfl.com/ecg-in-chronic-obstructive-pulmonary-disease/>

Ultrasound

Before intubation: COPD clip (A lines)

After intubation: Pneumothorax clip (no lung sliding)

Severe COPD Exacerbation with Pneumothorax

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

Sample Debriefing questions:

Clinical Debrief

1. What are your options for optimizing oxygenation in the agitated hypoxic patient? How do you optimize pre-oxygenation if your patient requires intubation (DSI)?
2. How do you trouble shoot high airway pressures, and what are the differentials to consider. How do we clinically apply the DOPE mnemonic?
3. In a patient who arrests after high airway pressures on a vent, how do your priorities change vs. a non-intubated PEA arrest patient (i.e. have a high suspicion for tension pneumothorax & have low threshold to decompress chest).

CRM Debrief

1. What techniques can be used to maintain situational awareness when critical procedures are required during a resuscitation (i.e. intubation, chest decompression).
2. What strategies can be used to maintain control in a chaotic resuscitation with an agitated patient?

Key Moments:

- Approach to/ use of DSI to optimize pre-oxygenation for emergent intubation
- Intubation of a hypoxic patient
- Recognition of increased airway pressures, and pneumothorax post intubation

References

1. <https://lifeinthefastlane.com/ccp/delayed-sequence-intubation/>
2. <https://emcrit.org/preoxygenation/>
3. <https://lifeinthefastlane.com/ccp/high-airway-and-alveolar-pressures/>

